

Review: Product of Inertia by Integration

1. Common definition: $I_{xy} = \int_A \tilde{x} \tilde{y} dA$

2. Remember: I_{xy} is relative to a pair of $\perp$ axes, so there is no " I_{base} " for I_{xy} ,

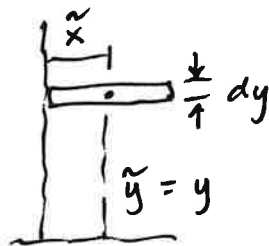
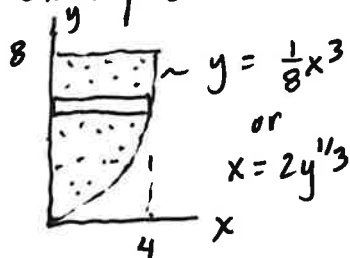
3. Can write $I_{xy} = \int_A dI_{xy}$, where $dI_{xy} = d\bar{I}_{x'y'} + dA \cdot \tilde{x} \tilde{y}$

but since virtually all elements are $\square$, $\square$, $\square$, $d\bar{I}_{x'y'} = 0$ for all of these.

So $I_{xy} = \int_A dI_{xy} = \int_A [0 + dA \tilde{x} \tilde{y}]$

$I_{xy} = \int_A \tilde{x} \tilde{y} dA \sim$ for any element.

4. Example:



$$dA = x \cdot dy$$

$$= (2y^{1/3}) dy$$

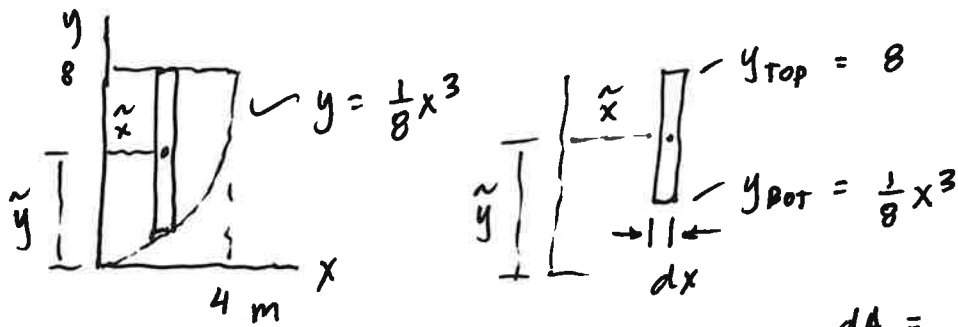
$$\tilde{x} = \frac{x}{2} = \frac{1}{2}(2y^{1/3})$$
$$= y^{1/3}$$

$$I_{xy} = \int_A \tilde{x} \tilde{y} dA$$

$$I_{xy} = \int_0^8 \underbrace{(y^{1/3})}_{\tilde{x}} \underbrace{(y)}_{\tilde{y}} \underbrace{(2y^{1/3})}_{dA} dy = \int_0^8 2y^{5/3} dy$$

$$= \frac{3}{8} \cdot 2 y^{8/3} \Big|_0^8 = \boxed{192 \text{ m}^4 = I_{xy}}$$

5. Next page, same problem, vertical element.



$$dA = (8 - \frac{1}{8}x^3) dx$$

$$\tilde{x} = x$$

$$\tilde{y} = \frac{1}{2} \left(8 + \frac{1}{8}x^3 \right)$$

$$I_{xy} = \int_A \tilde{x} \tilde{y} dA = \int_0^4 \underbrace{x}_{\tilde{x}} \underbrace{\left(\frac{1}{2} \left(8 + \frac{1}{8}x^3 \right) \right)}_{\tilde{y}} \underbrace{\left(8 - \frac{1}{8}x^3 \right)}_{dA} dx$$

$$\left(8 + \frac{1}{8}x^3 \right) \left(8 - \frac{1}{8}x^3 \right) = \left(64 - \frac{1}{64}x^6 \right)$$

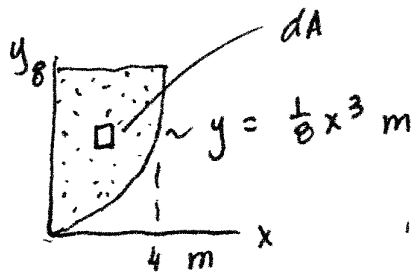
$$x[] = \left(64x - \frac{1}{64}x^7 \right)$$

$$I_{xy} = \int_0^4 \frac{1}{2} \left(64x - \frac{1}{64}x^7 \right) dx$$

$$= \frac{1}{2} \left[32x^2 - \frac{1}{8 \cdot 64}x^8 \right]_0^4$$

$$\boxed{I_{xy} = 192 \text{ m}^4}$$

Double Integral



Use a double integral:

$$I_{xy} = \int_A xy \, dA$$

See comments below.

$$= \int_0^4 \int_{\frac{1}{8}x^3}^8 xy \, dy \, dx$$

TI-89:

$$\begin{aligned} & \int \left(\int (x*y, y, (1/8)*x^3, 8), x, 0, 4 \right) \\ &= 192 \, m^4 \end{aligned}$$

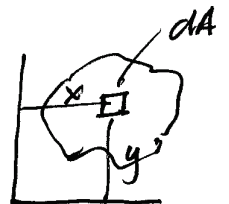
Note: I gave the general formula as

$$I_{xy} = \int_A \tilde{x} \tilde{y} \, dA$$

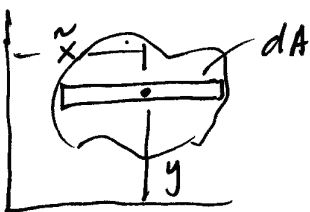
Hibbeler on his crib sheet gives the formula:

$$I_{xy} = \int_A xy \, dA$$

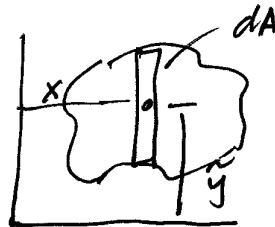
This is only true for a double integral:



For other choices of dA



or



$$\begin{aligned} dA &= dx \, dy \\ \text{or} &= dy \, dx \end{aligned}$$

One of the two distances will be x or y ... but the other will be $\tilde{y}$ or $\tilde{x}$.

$$\tilde{x} = \left(\frac{x_R + x_L}{2} \right), \quad \tilde{y} = \left(\frac{y_{BOT} + y_{TOP}}{2} \right)$$